

GEORGIA INSTITUTE OF TECHNOLOGY
COE3001-H MECHANICS OF DEFORMABLE BODIES

Midterm exam - 2025 Fall

Oct. 2, 9:30-10:45 am

Name: _____ GTID: _____

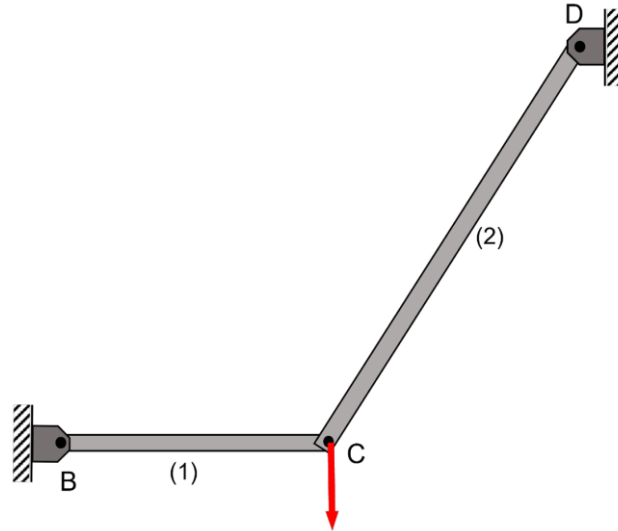
Please read the following instructions

- ✧ You have **75** minutes to complete this exam.
- ✧ You may use a single, double-sided, letter sized formula sheet.
- ✧ This question booklet contains **6** questions, **10** pages (including the cover) for a total of **100** points. Check to see if any pages are missing.
- ✧ DO NOT scribble or do rough work or make any stray marks on it. Use separate sheets for rough work.
- ✧ All the questions are compulsory, and all the notations have their usual meaning.
- ✧ Show all necessary steps for full credit. Clearly indicate your final answers.

Problem 1 (15 points):

Consider the truss above that is made up of elements (1) and (2).

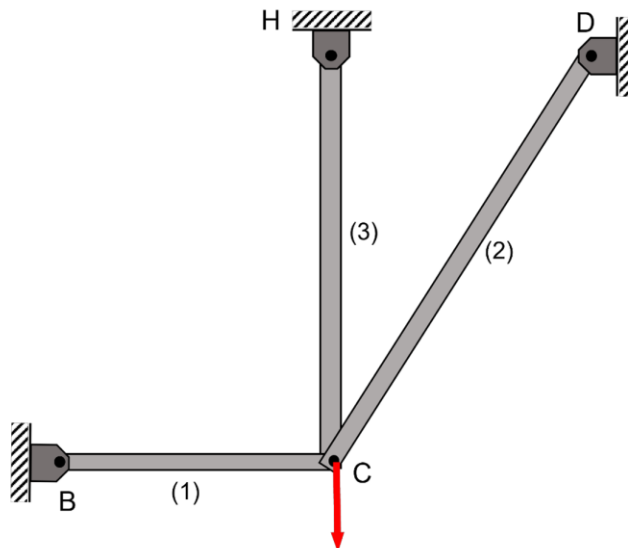
TRUE or FALSE: The stress in element (1) depends on the material of element (2).



Problem 2 (15 points):

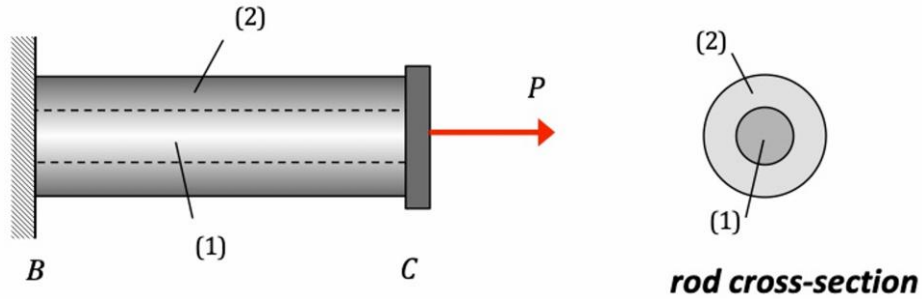
Consider the truss above that is made up of elements (1), (2) and (3).

TRUE or FALSE: The stress in element (1) depends on the material of (2) and (3).



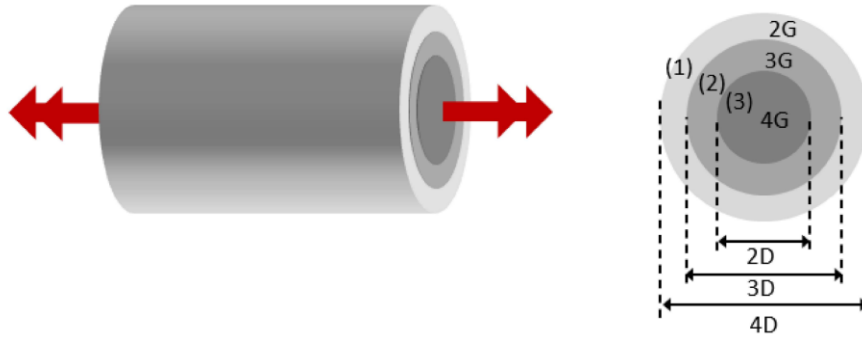
Problem 3 (15 points):

A rod is made of a shell (2) and core (1), where the areas are known to be $A_2 = 2A_1$ and the Young's modulus are known to be $E_1 = 3E_2$. The axial load P is acting on the rigid connector C. Let F_1 and F_2 be the axial loads carried by elements (1) and (2), respectively. Which one has a larger magnitude, $|F_1|$ or $|F_2|$?



Problem 4 (15 points):

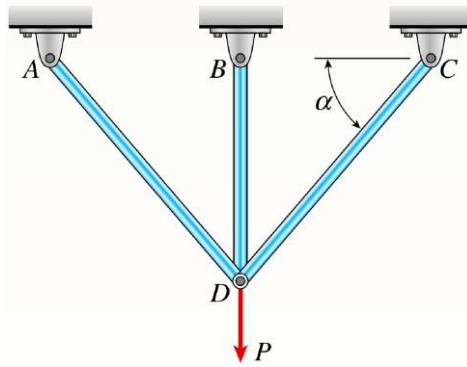
A torsion member consists of three concentric materials with different shear modulus as shown in the figure below. When a torsional loading is applied, there is no relative sliding. Which material experiences the largest shear stress?



Problem 5 (20 points):

A symmetrical framework consists of three pin-connected cylindrical bars, as shown in the figure. The structure is subjected to a vertical load $P = 137.3 \text{ kN}$ applied at joint D. The two inclined bars, AD and DC, are made of steel with Young's modulus $E_s = 210 \text{ GPa}$. The vertical bar BD is made of aluminum with Young's modulus $E_a = 70 \text{ GPa}$. The angle between each inclined bar and the horizontal is $\alpha = 45^\circ$. The length of bar BD is $L = 1 \text{ m}$. All bars have the same circular cross section with radius $r = 1.0 \text{ cm}$. Determine the vertical displacement of point D, δ . (Hint:

$$\sqrt{L^2 + (L + \delta)^2} \approx \sqrt{2}L + \frac{\delta}{\sqrt{2}} \text{ when } \delta \text{ is small})$$



Problem 6 (20 points):

A solid circular bar of diameter $d_1 = 25$ mm is enclosed by a tube of inner diameter $d_2 = 30$ mm and outer diameter $d_3 = 40$ mm. Both bar and tube are held rigidly by a support at end A and joined securely to a rigid plate at end B. The composite bar, which has a length L , is twisted by a torque $T = 400$ N · m acting on the end plate. The bar and tube have shear modulus of $G_1 = 200$ GPa and $G_2 = 100$ GPa, respectively. Calculate the maximum shear stress τ_1 and τ_2 in the bar and tube, respectively.

